MedScribe AI

Mobile-First Ambient AI Scribe
with Real-Time FHIR Conversion

Problem Statement: PS-1

Hackathon: HACKMATRIX 2.0

Organized by: Jilo Health × NJACK, IIT Patna

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PROPOSED SOLUTION

The Problem

India's healthcare system serves over 1.4 billion people, yet its clinical workforce remains critically understaffed. The doctor-to-patient ratio stands at roughly 1:1,000, well below the WHO recommendation of 1:1,000 in urban areas and far worse in rural settings. In this high-pressure environment, physicians spend an estimated **2–3 hours every day** on clinical documentation—writing case notes, filling discharge summaries, coding diagnoses, and preparing referral letters. This administrative burden directly reduces the time available for patient care, accelerates burnout, and contributes to India's growing physician attrition crisis.

Compounding the problem is India's unique linguistic landscape. Doctor-patient conversations routinely occur in **Hindi-English code-mixed speech**—a patient might describe their symptoms as *"pet mein dard ho raha hai, 3 din se fever bhi hai"* while the doctor responds with clinical English interspersed with Hindi reassurances. Global ambient scribe tools (Abridge, Nuance DAX, Suki) are built for monolingual English clinical settings and **fail catastrophically** on code-mixed Indian conversations. Meanwhile, India's ambitious **Ayushman Bharat Digital Mission (ABDM)** mandates FHIR-compliant health records for interoperability, yet no affordable tool exists that can convert a spoken consultation into standards-compliant digital records. Rural telemedicine expansion—a national priority—remains bottlenecked by the documentation overhead imposed on already-overstretched physicians.

Our Solution

MedScribe AI is a mobile-first Progressive Web App that transforms spoken doctor-patient conversations into structured clinical notes and FHIR R4–compliant resources in real time. The workflow is radically simple: a doctor opens the app on their phone, taps a single record button, and speaks naturally with their patient in any mix of Hindi and English. As the conversation flows, MedScribe AI streams a live transcript while simultaneously extracting chief complaints, diagnoses with ICD-10 codes, medications with dosage and RxNorm mappings, vitals coded to LOINC, allergies, and a structured SOAP note—all updated incrementally on screen.

What sets MedScribe AI apart is the integration of **Clinical Decision Support** directly into the documentation pipeline. As medications and allergies are extracted, the system checks for drug-drug interactions, allergy cross-reactivity, and dosage safety—raising real-time alerts before the prescription is finalized. A **Differential Diagnosis Engine** suggests alternative diagnoses with supporting evidence and distinguishing tests. Every consultation produces a scored FHIR R4 Bundle (Grade A–D) ready for ABDM integration, complete with Patient, Encounter, Condition, Observation, MedicationRequest, AllergyIntolerance, CarePlan, and ServiceRequest resources. The system learns continuously: when a doctor corrects an extraction, that correction is stored and injected as a few-shot example into future calls, improving accuracy without model retraining. **Zero API cost for speech recognition**—we leverage the browser's native Web Speech API—makes this viable even in resource-constrained settings.

System Architecture

Data Flow

The browser captures speech via the Web Speech API (supporting Hindi + English), streams recognized text over a WebSocket to the FastAPI backend. Accumulated transcript segments are batched every 10–15 seconds and sent to Gemini 2.5 Flash with a structured clinical extraction prompt. The AI returns structured JSON which is mapped to FHIR R4 resources, scored for completeness, and run through the CDS engine for safety checks. All results stream back to the frontend over the same WebSocket connection.

Technology Stack

Layer	Technology	Rationale
Frontend	React 18, Vite, Tailwind CSS	PWA-capable, fast HMR, mobile-first utility CSS
Speech-to-Text	Web Speech API (browser-native)	Zero API cost, Hindi-English support, no latency
Backend	Python FastAPI + WebSocket	Async-native, auto-docs, WebSocket first-class
AI Extraction	Google Gemini 2.5 Flash	Fast structured output, code-mixed understanding
FHIR Mapping	Custom Python + Pydantic models	Type-safe R4 resource generation with validation
CDS Engine	Rule-based Python service	Deterministic safety checks, no AI hallucination risk
Terminology	ICD-10, SNOMED CT, RxNorm, LOINC	International coding standards, ABDM alignment

Security & Privacy Considerations

- Speech processed in-browser—raw audio never leaves the device
- WebSocket connection scoped to session; no persistent patient data stored on server
- FHIR resources generated on-demand, exportable as local JSON files
- No patient identifiers stored in correction/learning data
- PWA supports offline-capable UI shell via service worker caching

Key Features

FEATURE 01

Hindi-English Code-Mixed Understanding

Natively handles conversations where doctors and patients switch freely between Hindi and English. Extracts clinical entities from phrases like *"pet mein dard"* (abdominal pain) and maps Indian drug brand names (Crocin, Combiflam) to their generic equivalents with correct RxNorm codes.

FEATURE 02

Clinical Decision Support (CDS)

Real-time safety engine with **10 drug-drug interaction rules**, **3 allergy cross-reactivity checks**, and **10 dosage validation rules**. Catches dangerous combinations (e.g., Warfarin + Aspirin, NSAID in renal impairment) before the prescription is finalized. Severity-graded alerts (high / moderate / low).

FEATURE 03

FHIR R4 + ABDM Compliance

Generates a complete FHIR R4 Bundle with **8 resource types**: Patient, Encounter, Condition (ICD-10), Observation (LOINC), MedicationRequest (RxNorm), AllergyIntolerance, CarePlan, and ServiceRequest. Each resource is quality-scored (Grade A–D) for coding completeness and inter-resource referential integrity.

FEATURE 04

Differential Diagnosis Engine

For every primary diagnosis, the AI suggests **2–3 alternative diagnoses** with supporting evidence extracted from the conversation and recommended distinguishing tests. Helps doctors consider conditions they might overlook, supporting diagnostic rigor and reducing misdiagnosis rates.

FEATURE 05

Continuous Learning from Corrections

Doctors can edit any extracted clinical note field and click **"Save & Teach AI."** Corrections are stored and injected as few-shot examples into future Gemini extraction calls. The system improves with use—no model fine-tuning or retraining required. Accuracy compounds across consultations.

FEATURE 06

Export Suite & Demo Mode

One-click export to **print-ready clinical notes** (opens browser print dialog), **FHIR Bundle JSON download**, and **clipboard copy**. Built-in demo mode with 3 pre-loaded scenarios (Viral Fever, Diabetes Follow-Up, Cardiac + Safety Alerts) ensures reliable presentations even without live microphone access.

Additional Capabilities

Capability	Description
FHIR Quality Scoring	Automated Grade A–D scoring checking resource completeness, coding systems, and cross-references
Risk Factors & Tests	AI identifies patient risk factors and suggests appropriate diagnostic tests
Dark Mode	Full dark theme support for low-light clinical environments (night shifts, ICU)
PWA Install	Installable on mobile home screen; offline UI shell via service worker
Real-Time Streaming	WebSocket-based incremental updates; transcript + note + FHIR update simultaneously
Toast Notifications	Non-blocking alerts for CDS warnings, extraction status, and connection state

Competitive Differentiation

Capability	MedScribe AI	Abridge	Nuance DAX	Suki AI
Hindi-English Code-Mixed	✔ Native	✘ English only	✘ English only	✘ English only
ABDM / FHIR R4 Alignment	✔ Full R4 Bundle	● Partial	● Via Epic	● Partial
Clinical Decision Support	✔ Built-in CDS	✘ None	✘ External	✘ None
Indian Drug Brand Mapping	✔ Crocin → Paracetamol	✘	✘	✘
Differential Diagnosis	✔ With evidence	✘	✘	● Basic
Speech API Cost	✔ $0 (browser-native)	Proprietary	Proprietary	Proprietary
Continuous Learning	✔ Few-shot injection	✘	✘	✘
Pricing (India)	Open source / Free	~$1,000/mo	~$1,500/mo	~$400/mo

Impact Metrics

13×
FASTER DOCUMENTATION

$0
SPEECH API COST

23
SAFETY RULES (CDS ENGINE)

8
FHIR R4 RESOURCE TYPES

Projected Clinical Impact

- **Time savings:** Reduces documentation time from ~15 minutes to ~1 minute per consultation, freeing 2+ hours/day for direct patient care.
- **Patient safety:** Real-time CDS catches dangerous drug interactions and allergy cross-reactivity before prescriptions are finalized—errors that manual review often misses under time pressure.
- **Digital health readiness:** Automatic FHIR R4 output means every consultation is ABDM-ready from day one, accelerating India's health record interoperability goals.
- **Equity:** Browser-native speech recognition with zero API cost makes ambient scribing accessible to small clinics and rural telemedicine setups, not just well-funded hospital chains.
- **Diagnostic quality:** Differential diagnosis suggestions and risk factor identification support clinical reasoning, especially for junior doctors and GPs handling unfamiliar presentations.

Live Demo

Available Demo Scenarios

#	Scenario	Highlights
1	Viral Fever (Hindi-English)	Code-mixed transcript, Crocin → Paracetamol mapping, ICD-10 coding
2	Diabetes Follow-Up	Medication review, HbA1c observation, CarePlan with diet instructions
3	Cardiac + Safety Alerts	Warfarin + Aspirin interaction, allergy cross-reactivity, high-severity CDS
4	Live Microphone	Real-time recording with actual doctor-patient conversation

How to Run

```
# Clone and install
$ git clone https://github.com/your-org/medscribe-ai.git
$ cd medscribe-ai
$ npm install

# Set Gemini API key
$ echo "GEMINI_API_KEY=your_key" > backend/.env

# Start both frontend and backend
$ npm run dev

# Open http://localhost:5173 on your phone or browser
```

Future Roadmap

PHASE 1 — NOW

Hackathon MVP

Hindi-English ambient scribe, FHIR R4 output, CDS engine, differential diagnosis, demo mode, PWA.

PHASE 2 — Q2 2026

Regional Language Expansion

Add Tamil, Telugu, Bengali, Marathi support. Fine-tune extraction for regional medical terminology and prescription patterns.

PHASE 3 — Q3 2026

EHR Integration & ABDM Bridge

Direct integration with Indian EHR systems (Bahmni, OpenMRS). ABDM Health Information Provider (HIP) module for national health record exchange.

PHASE 4 — Q4 2026

Edge Deployment & Offline Mode

On-device AI inference for areas with unreliable internet. Local Whisper model for offline speech recognition. Full offline-first architecture for rural deployment.

Acknowledgments

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